

NLP Project Requirements Document

AI Assistant for Accessing Public Health Information

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1 Introduction

Access to reliable health information is a major societal challenge. Many citizens struggle to understand medical recommendations, vaccination guidelines, or administrative procedures related to healthcare services. These difficulties are often due to complex language, scattered information sources, or limited digital literacy.

Natural Language Processing (NLP) offers powerful tools to make health information more accessible. This project proposes the development of an **AI-powered assistant that helps users find and understand public health information using natural language queries**.

The system will allow users to ask questions in plain language and receive clear answers extracted from trusted public health documents.

This project will strengthen the student's skills in:

- Natural Language Processing
- Information retrieval
- Large Language Model orchestration
- Real-world AI system design

2 Project Objectives

2.1 General Objective

Develop a simple NLP-based assistant capable of answering user questions about public health guidelines using official documents.

2.2 SMART Objectives

- **Specific:** Build a question-answering system that extracts information from public health documents.
- **Measurable:** Achieve at least **70% correct answers** on a small evaluation dataset of health-related questions.
- **Achievable:** Use existing NLP libraries (spaCy, LangChain) and pre-trained models rather than training models from scratch.
- **Realistic:** Work with a limited dataset (10–30 public health documents).
- **Time-bound:** Complete the project within **6 weeks**.

3 Problem Statement

Many official healthcare documents contain valuable information but are difficult to navigate. Citizens often ask questions such as:

- “What vaccines are recommended for children?”
- “What are the symptoms of influenza?”
- “Where can I get tested for COVID-19?”

Finding answers requires reading multiple documents, which is time-consuming.

The goal of this project is to build a **Natural Language Question Answering System** capable of retrieving relevant information from official health resources and presenting clear answers.

4 Scope of the Project

4.1 Included Features

- Processing public health documents (PDF or text)
- Extracting and indexing information
- Accepting natural language questions
- Retrieving relevant document passages
- Generating short answers

4.2 Excluded Features

- Medical diagnosis
- Real-time hospital databases
- Training large models from scratch
- Multilingual support beyond English (optional extension)

5 System Architecture

The system will follow a **Retrieval-Augmented Generation (RAG)** architecture.

5.1 Pipeline Overview

1. Document collection
2. Text preprocessing
3. Information extraction
4. Vector indexing
5. Question processing
6. Document retrieval
7. Answer generation

6 Technologies and Tools

6.1 Core NLP Tools

- **spaCy**

Used for:

- Tokenization
- Named Entity Recognition (NER)
- Text preprocessing

Justification: spaCy is fast, production-ready, and beginner-friendly.

- **LangChain**

Used for:

- Building the RAG pipeline
- Connecting LLMs with document retrieval
- Managing prompt templates

Justification: LangChain simplifies the development of LLM-based applications.

6.2 Additional Recommended Tools

Tool	Type	Justification
Python	Programming language	Standard language for NLP and data science.
Sentence Transformers	Embedding model	Converts text into semantic vectors for similarity search.
FAISS or ChromaDB	Vector database	Stores embeddings and allows fast similarity search.
HuggingFace Transformers	NLP models	Provides pre-trained models for question answering and embeddings.
Streamlit	Web interface	Allows creation of a simple user interface for testing the assistant.

7 Dataset

The student will collect a small dataset of public health documents.

Possible sources:

- WHO health guidelines
- Government health websites
- Public vaccination guides

7.1 Dataset Requirements

- 10–30 documents
- Text or PDF format
- Trusted sources only

8 Functional Requirements

FR1 The system must ingest health documents.

FR2 The system must preprocess text using spaCy.

FR3 The system must generate embeddings for document chunks.

FR4 The system must store embeddings in a vector database.

FR5 The system must accept natural language questions from users.

FR6 The system must retrieve the most relevant document passages.

FR7 The system must generate a clear answer based on retrieved context.

FR8 The system must display the answer and the source document.

9 Non-Functional Requirements

NFR1 The system must respond to queries in less than 5 seconds.

NFR2 The interface must be simple and accessible.

NFR3 The system must rely only on publicly available tools.

NFR4 The project must be reproducible.

10 Evaluation Metrics

The system will be evaluated using the following metrics:

- **Answer Accuracy**
- **Retrieval Precision**
- **User Satisfaction (manual testing)**

Example evaluation process:

- Prepare 20 test questions
- Measure how many answers are correct
- Calculate accuracy

11 Project Timeline

Week	Tasks
Week 1	Literature review and dataset collection
Week 2	Document preprocessing with spaCy
Week 3	Embeddings and vector database setup
Week 4	Implement RAG pipeline using LangChain
Week 5	Build a simple Streamlit interface
Week 6	Testing, evaluation, and final report

12 Expected Deliverables

- Source code repository (GitHub)
- Dataset of collected health documents
- Running prototype
- Technical report
- Demonstration of the system

13 Possible Extensions

If time permits, the student may explore:

- Multilingual support
- Improved entity recognition for medical terms
- Conversation memory
- Integration with speech input

14 Conclusion

This project provides a realistic introduction to modern NLP systems by combining traditional NLP tools and Large Language Models.

By implementing a health information assistant, the student will gain practical experience in:

- NLP pipelines
- Information retrieval
- Retrieval-Augmented Generation
- AI system deployment

At the same time, the project addresses a meaningful societal problem: improving access to reliable health information.